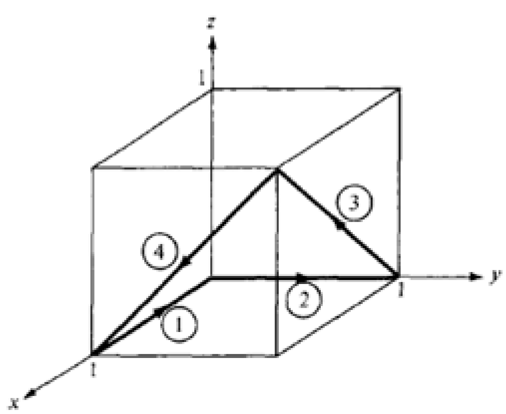
Princess Sumaya University for Technology
Communication Engineering Department
Applied Electromagnetics
Worksheet 3

1. Given that $\boldsymbol{F} = x^2\boldsymbol{a_x} - xz\boldsymbol{a_y} - y^2\boldsymbol{a_z}$, calculate the circulation of $\boldsymbol{F}$ around the (closed) path shown in the figure below:



2. Find the gradient of these scalar fields:
   a. $\mathrm{U} = 4xz^2 + 3yz$
   b. $\mathrm{W} = 2\rho(z^2 + 1)\cos\phi$
   c. $\mathrm{H} = r^2 \cos\theta \cos\phi$

3. Find the divergence and curl of the following vectors:
   a. $\boldsymbol{A} = e^{xy}\,\boldsymbol{a_x} + \sin xy\,\boldsymbol{a_y} + cos\,^2xz\,\boldsymbol{a_z}$
   b. $\boldsymbol{B} = \rho z^2 \cos\phi\,\boldsymbol{a}_\rho + zsin\,^2\phi\,\boldsymbol{a_z}$
   c. $\boldsymbol{C} = r\cos\theta\,\boldsymbol{a}_r - \frac{1}{r}\sin\theta\,\boldsymbol{a}_\theta + 2r^2\sin\theta\,\boldsymbol{a}_\phi$